SIH26189

AI-Powered Criminal Network Analysis System

Synthetic Sample Data for Project Demo

Important: All records in this document are fictional/synthetic and intended only for software development, testing and SIH demonstration.

1. Sample FIR — fir_001.txt

```
FIRST INFORMATION REPORT

FIR ID: FIR-2026-001
Date: 12-08-2026
Police Station: Central Cyber Cell

The complainant reported suspicious financial activity involving
Rahul Sharma and Amit Kumar.

During preliminary investigation, Rahul Sharma was found associated
with mobile number 9876543210. The number communicated with
9123456789 on multiple occasions.

A bank account ending in 7890 was allegedly used for receiving
■5,00,000 from another account.

Amit Kumar was identified as a person associated with mobile
number 9123456789.

Vehicle UP32AB1234 was also found associated with Rahul Sharma.

Investigation is ongoing.
```

Expected extracted entities

```
PERSON: Rahul Sharma, Amit Kumar
PHONE: 9876543210, 9123456789
BANK ACCOUNT: XXXXXX7890
AMOUNT: ■5,00,000
VEHICLE: UP32AB1234
FIR: FIR-2026-001
```

2. CDR — cdr.csv

```
call_id,caller,receiver,date,time,duration_seconds
C001,9876543210,9123456789,2026-08-05,10:32,120
C002,9123456789,9988776655,2026-08-05,11:15,245
C003,9876543210,9988776655,2026-08-06,09:42,180
C004,9988776655,8877665544,2026-08-06,14:20,95
C005,9123456789,8877665544,2026-08-07,16:45,310
C006,8877665544,7766554433,2026-08-08,18:10,150
C007,9876543210,9123456789,2026-08-09,12:05,420
C008,7766554433,9988776655,2026-08-09,13:22,75
C009,9123456789,9988776655,2026-08-10,15:30,280
C010,9988776655,8877665544,2026-08-11,17:55,360
```

3. Bank Transactions — bank_transactions.csv

```
transaction_id,from_account,to_account,amount,date,time,reference
T001,ACC1001,ACC2002,500000,2026-08-05,12:10,TXN-A001
T002,ACC2002,ACC3003,300000,2026-08-06,14:35,TXN-A002
T003,ACC3003,ACC4004,180000,2026-08-07,16:20,TXN-A003
T004,ACC4004,ACC5005,120000,2026-08-08,18:45,TXN-A004
T005,ACC2002,ACC6006,75000,2026-08-09,11:15,TXN-A005
T006,ACC6006,ACC7007,50000,2026-08-10,13:40,TXN-A006
T007,ACC3003,ACC8008,90000,2026-08-11,15:10,TXN-A007
T008,ACC1001,ACC9009,25000,2026-08-12,10:30,TXN-A008
```

4. Person–Entity Mapping — persons.csv

```
person_id,name,phone,bank_account,vehicle,email
P001,Rahul Sharma,9876543210,ACC1001,UP32AB1234,rahul.demo@example.com
P002,Amit Kumar,9123456789,ACC2002,UP32CD5678,amit.demo@example.com
P003,Rohit Verma,9988776655,ACC3003,UP32EF9012,rohit.demo@example.com
```

```
P004,Vikas Singh,8877665544,ACC4004,UP32GH3456,vikas.demo@example.com
P005,Neeraj Gupta,7766554433,ACC5005,UP32JK7890,neeraj.demo@example.com
```

5. IP Logs — ip_logs.csv

```
log_id,phone,ip_address,date,time
IP001,9876543210,203.0.113.10,2026-08-05,10:30
IP002,9123456789,203.0.113.11,2026-08-05,11:12
IP003,9988776655,203.0.113.12,2026-08-06,09:40
IP004,8877665544,203.0.113.13,2026-08-07,16:40
IP005,7766554433,203.0.113.14,2026-08-08,18:05
```

6. Vehicle Records — vehicles.csv

```
vehicle_id,registration,owner,location,date
V001,UP32AB1234,Rahul Sharma,Lucknow,2026-08-05
V002,UP32CD5678,Amit Kumar,Lucknow,2026-08-06
V003,UP32EF9012,Rohit Verma,Kanpur,2026-08-07
V004,UP32GH3456,Vikas Singh,Kanpur,2026-08-08
V005,UP32JK7890,Neeraj Gupta,Lucknow,2026-08-09
```

7. Expected Graph Relationships

```
Rahul Sharma
■■■ USES ■■> 9876543210
■■■ OWNS ■■> ACC1001
■■■ DRIVES ■■> UP32AB1234

9876543210
■■■ CALLS ■■> 9123456789
■■■ USED BY ■■> Amit Kumar
■■■ OWNS ■■> ACC2002
■■■ TRANSFERRED TO
↓
ACC3003
↓
ACC4004
↓
ACC5005
```

8. Demo Investigation Flow

```
LOGIN
↓
CREATE CASE
↓
UPLOAD FIR PDF
↓
PDF → TEXT
↓
ENTITY EXTRACTION
↓
INVESTIGATOR REVIEW
↓
UPLOAD CDR CSV
↓
UPLOAD BANK CSV
↓
ENTITY RESOLUTION
↓
BUILD NEO4J GRAPH
↓
SEARCH: Rahul Sharma
↓
3-HOP / 4-HOP GRAPH
↓
MONEY TRAIL
↓
CENTRALITY / NETWORK ANALYSIS
↓
TIMELINE
↓
GENERATE REPORT
```

9. Expected Demo Queries

```
1. Find all entities connected to Rahul Sharma.
2. Show Rahul Sharma's phone numbers.
3. Show all calls from 9876543210.
4. Find a 3-hop path from Rahul to Amit.
5. Show the money trail starting from ACC1001.
6. Find accounts receiving more than ■1,00,000.
7. Calculate high-degree communication nodes.
8. Show all entities mentioned in FIR-2026-001.
9. Display events in chronological order.
10. Generate an investigation summary report.
```

10. Suggested Neo4j Node Types

```
Person
Phone
BankAccount
Transaction
IP
Vehicle
FIR
Location
CryptoWallet
Email
Organization
```

11. Suggested Neo4j Relationships

```
(:Person)-[:USES]->(:Phone)
(:Person)-[:OWNS]->(:BankAccount)
(:Person)-[:DRIVES]->(:Vehicle)
(:Person)-[:USES]->(:IP)
(:Person)-[:OWNS]->(:CryptoWallet)
(:Phone)-[:CALLED]->(:Phone)
(:BankAccount)-[:TRANSFERRED_TO]->(:BankAccount)
(:Person)-[:MENTIONED_IN]->(:FIR)
(:Person)-[:LOCATED_AT]->(:Location)
```

Safety / interpretation note: This dataset is fictional. Graph centrality, anomaly scores or AI outputs should be presented as investigative leads requiring human verification, not as proof of criminal activity.